



Flying Lifteraft

Technical Direction



Si Wing Foster Ng
CID: 01716908

Group Design Project
H401 MEng Aeronautical Engineering

13/06/2022

FLR22

Subteam	Members
Technical Directors	Foster Ng
	Julian Fung
	Daniel Earl
Aerodynamics	Anthony Haiser
	Kwok Leung Yim
	Marius Borsu Lefevre
Structures	Sorawis Itthithavorn
	Antoine Rieuneau
	Olivier Sussfeld
Propulsion and Systems	Jordi Frittoli
	Matan Hamburger
	Wei Han Lee
Performance, Stability & Control	Natalia Melo Fernandez
	Farsai Sittichoksuwan
	Kristoff Ahlner
Simulation	Joshua Binns
	Elia Chimonides
	Luis Ferreira da Silva Marques
Simulation	Yi En Puah
	Bojun Vito Yang
	Gabisanth Seyon
Simulation	Rohit Singh

Abstract

This report presents part of the technical direction of the FLR22 project. Major design decisions and the justifications behind them are discussed. The EASA UAV regulations, relevant to the system in the target market, are explored. A detailed cost estimation for the program was produced, and the expected lifespan of the drone calculated - the total cost for 100 aircraft is expected to be 15 million GBP, with an 80 mission expectancy per drone and a 2250 GBP cost per sortie. Finally, the most efficient and accurate payload drop method for the UAV was determined to be a low-speed drop using the quad-rotors (below the stall speed), achieving an accuracy of 5 m from a drop height of 18 m with a ground speed of up to 10 m/s.

Contents

List of Figures	iv
List of Tables	v
List of Symbols	vi
1 Introduction	1
2 Team Management & Conflict Resolution	1
3 Regulations	4
4 Cost Estimation and Lifespan	5
4.1 Parts	7
4.2 Manufacturing	7
4.3 Mission-specific	8
5 Payload Drop Method Trade-off Analysis	8
5.1 Evaluation	12
6 Conclusion	12
References	a
Appendix	b

List of Figures

1	Migrant Casemap in the Mediterranean [1]	5
2	Cost vs. Quantity	6
3	Method 3	10
4	Method 4	10
5	Method 1	11
6	Method 2	11
7	Gantt Chart excerpt	b
8	Work Breakdown excerpt	b

List of Tables

1	Part costs	7
2	Manufacturing costs	7
3	Mission-specific costs	8
4	Payload Drag Coefficients	9

List of Symbols

FLR22	Flying Liferafts 2022
FAT	Functional Area Team
MS	Microsoft
ICE	Internal Combustion Engine
MTOM	Maximum Take-Off Mass
VTOL	Vertical Take-Off and Landing
UAS	Unmanned Aircraft System
UAV	Unmanned Aerial Vehicle
x_{CG}	x-position of the Centre of Gravity of the aircraft
RPM	Revolutions Per Minute
$C_{L_{Max}}$	Maximum lift coefficient
MSA	Minimum Safe Altitude
EASA	European Union Aviation Safety Agency
ADS-B	Automatic Dependent Surveillance-Broadcast (a type of transponder)
EU	European Union
CS-23	Certification Specification 23
GBP	Great British Pounds
APU	Auxiliary Power Unit
D	Drag force
C_D	Drag coefficient
ρ	Air density
U	True Air Speed
A	Cross-sectional area
ETL	Effective Translational Lift

1 Introduction

The aim of the FLR22 Group Design Project was to produce a cost-effective solution to aid a person stranded at sea, maximising their survivability until surface vessels arrived. Compactness, ease of use and all-weather operations were emphasised as requirements in the design brief.

Personally, an oversight role was taken in the project, with major design decisions and conflicts being resolved throughout. Regulations for the aircraft were also explored, and a comprehensive cost estimation produced to present a workable business case for the product. Finally, a technical analysis on the payload drop method was performed to determine the most accurate and efficient way of delivering aid.

2 Team Management & Conflict Resolution

At the outset of the project, the first goal was to determine the mission parameters and to explore as much of the design space as possible. In order to do this, the first week was spent reviewing literature and brainstorming different baseline configurations in four separate subgroups. Each subgroup was formed by allocating one person from each functional area team (FAT), ensuring that there were a mix of interests, and an approximate timeline for the week was set out. It was decided that all subgroups would work completely separately to each other, attempting to limit idea overlap between groups. Midweek, a short summary of each idea was also obtained from the subgroups to ensure that each of them had unique points. The end of each work week was defined to be the weekly project meeting, occurring every Tuesday morning.

Furthermore, both technical directors also used this time to establish a Gantt chart (Figure 7 in the Appendix) for the remaining four weeks of the project, create feedback forms for the supervisors, as well as a list of evaluation metrics for the different baseline configurations, shown below:

- Cost
- Range
- Speed/Time to target
- Ease of use
- Adaptability
- Land/Sea deployable or both?
- Size
- Robustness
- Effectiveness (number of people saved per kg of payload)

At the end of the work week, each group was given 15 minutes to present their idea. After obtaining feedback from the supervisors, a group discussion with the entire team was conducted (led by the other technical director), utilising empirical data to decide on the final mission parameters and the criteria set out previously to establish an initial team-wide baseline configuration. A maximum response time of 30 minutes was set to ensure that the rescuees would still be conscious in the worst-case scenario of 0-4° water [2]. A high wing was chosen to allow for a carry-through wing box structure, as well as minimising interference with the payload dropping mechanisms. A more detailed work package breakdown (Figure 8 in the Appendix) was created after the baseline meeting to keep track of individual tasks within each subteam.

In order to carry out the new team-wide initial sizing, another subgroup (comprised of several members from the previous two most relevant subgroups' designs) was formed. Values for major parameters here were either arbitrarily decided or taken from similar existing aircraft, with the expectation of major changes once some numerical analysis was completed. The main focus of the team in week 2 was to set up a framework and the code necessary for the design, allowing changes to be easily propagated through the system when made.

Throughout the design process, there were many conflicts in design choices for the aircraft based on differing technical interests. These were mediated based on the overall mission requirements

and trade-off analyses in each case. In many cases, a decision was postponed until a numerical justification could be provided for any/all of the options. A running tally of bottlenecks for each FAT was also kept verbally or through MS Teams, allowing each FAT to be rapidly re-prioritised (determined by various cross-dependencies for technical values) at each stage of the design.

The first major conflict resolved (with both technical directors) was the suggestion of a ducted fan configuration versus a conventional quad-rotor design. The key factors in the decision were the drag reduction provided by the removal of two external rotors (determined to be roughly 40% by the Aerodynamics subteam), against the increased weight of the internal combustion engine (an increase of 3.2 kg) and vastly reduced controllability owing to the difficulty of throttling an internal combustion engine (ICE) within a reasonable timeframe. The thrust vectoring method of rotating the entire ICE was also considered too slow; an alternate flap mechanism to avoid engine rotation was also suggested but was overly complex and not redundant. At the time, 3.2 kg represented over 11% of our first maximum takeoff mass (MTOM) of 28 kg, a significant increase - hence, with the emphasis on being lightweight to minimise VTOL power requirements, it was decided that the 40% drag reduction was not worth the trade-offs, especially considering that we were able to carry an excess of fuel in the aircraft.

Another major design decision was whether to include an alternator in the UAS. An alternator would have provided the drone with the capability to charge the batteries in flight through the ICE, allowing the battery pack to be downsized. However, Propulsion discovered that the smallest alternator that could feasibly charge the batteries in the amount of time required (20 minutes of cruise) would weigh 9 kg, again an inhibitive large weight increase. Moreover, the minimum battery size was limited by the power draw of the motors, rather than the energy requirement. Potential issues also arose with the packing of the system. A middle-ground option of a smaller alternator could be included, but this would purely be to avoid having to charge the UAV from the launch ship. The ICE would have needed to continue running on deck (with a clutch to decouple the pusher) to utilise the alternator (a safety hazard), and it would have been insufficiently quick to charge the batteries during flight. Carrying several kilos of extra weight that would only be utilised while not on mission was also considered extremely detrimental to the drone's ultimate goal - therefore, the entire idea was discarded by both technical directors and the requirement for deck charging was implemented. In order to mitigate against slow charging (in case the drone had to be rapidly relaunched), the batteries were designed to be hot-swappable. Rough sea states were a concern for this, as exposing the battery compartment would only be feasible in dry conditions, but the drone is compact enough at 2.2 m x 1.5 m that it should be able to fit through most doors/bulkheads for dry conditions indoors. Despite the weight penalty, a clutch was ultimately still included in the drone design to improve safety for the operator(s).

In the middle of week 3, it was discovered that the initial sizing and MTOM of 28 kg was inaccurate and insufficient for the necessary payload weight (9 kg) and VTOL requirements. In particular, the original battery mass fraction used was far too low - this was corrected by rubber-banding the actual battery size of 6 kg (provided by Propulsion) for a 28 kg drone. A new initial sizing with this new battery mass fraction of 21% was conducted, along with various members from each FAT, to determine the new MTOM. The aspect ratio was selected to match a maximum wingspan requirement (2 m at this point), which derived from the minimum deck size for launching (found by the other technical director to be 4 x 2.52 m), including 1 m of safety margin either side of the wings. Compactness was emphasised in the design philosophy. This also informed the maximum aircraft length of 1.5 m.

The stall speed was determined from the loiter speed, which originated from an image recognition constraint. A fixed-wing flight loiter was chosen rather than a hover loiter, as the aim was to minimise use of the electric motors to cut down as much as possible on battery mass. Aerodynamics would have preferred as high a stall speed as possible, but Control wanted it as low as possible for an easier transition from VTOL to fixed-wing flight and better image recognition at loiter speed. As the drone was required to be able to locate the target(s) on its own, the camera's frame rate and resolution became the limiting factors for loiter speed. Using the specifications of the given thermal camera, chosen by Propulsion and Systems, an initial maximum loiter height was chosen to be able to obtain 3 pixels of a 0.5 m-wide target; this came with a requisite frame rate for 99% + recognition accuracy which informed the chosen loiter speed of 40 m/s. This was then used, along

with the maximum horizontal gust requirement from CS-23 and a safety factor of 1.1, to determine the stall speed:

$$V_{Stall} = \frac{V_{Loiter} - 7.62}{1.1} = 29.4 \quad (1)$$

Although even lower stall speeds were considered, to further aid the transition phase, it was found that going lower would result in an inefficient wing with the geometry restrictions already in place. The final MTOM was determined to be 51.5 kg, with a wing loading of 832 N/m and thrust-to-weight ratio of 0.218.

There was a design choice for the pylons supporting the electric motors to be attached to the fuselage or the wings. The trade-offs were potentially shorter and lighter pylons, but complicated aerodynamic interactions with the wing, or structural issues from wing mounting points and higher weight penalties. In the end, with the other technical director and factoring precedent into consideration, wing-mounted pylons were chosen - the wing turned out to be over-designed anyway, negating any structural problems, and complicated aerodynamic/control complications were avoided. Moreover, the pylon positioning became independent of x_{CG} , circumventing future issues there.

The initial quad-rotor propeller design was found to be problematic - the rotor radius was nearly as large as the semi-span, needing a very narrow fuselage to meet the overall maximum wingspan requirement. As such, the decision was made to resize the propeller and increase the RPM of the electric motors to meet the thrust requirements instead. This resulted in a smaller propeller radius of 0.3 m which was ultimately less efficient, but allowed Structures to design a reasonable fuselage for packing and layout. As the final sizing still had the propellers extending 0.1 m beyond the wingspan on each side, the wingspan was ultimately extended to 2.2 m by extrapolating the existing geometry of the wing, an essentially “free” gain in lifting efficiency. The wing had already been over-designed structurally, so no problems arose there.

This change in wingspan aided Aerodynamics in recovering some lift characteristics, as their initial $C_{L_{Max}}$ estimate of 1.45 was inaccurate; it was actually closer to 1.2 as obtained from their Star-CCM+ simulations. Aerodynamics had already previously wanted to increase the wingspan through the use of foldable wings, but this had been vetoed for weight and complexity reasons. A change in twist to -4 degrees was also allowed as it did not affect any other subteam significantly. On the other hand, a increase in taper ratio, purported to move the stall point further inboard of the ailerons, was rejected, as it would have also caused the pylons with the rotors to move further out, negating the “free” gain in wingspan. It was also quite late on in the project, requiring significant redesign from multiple subteams if the change was to be propagated; thus time was also a prohibitive factor. Instead, the ailerons were moved to 45% span from 40%, solving the stall issue. The taper ratio was also effectively decreased, as the new wing was longer and therefore had a smaller tip chord compared to the previous iteration.

The tail position and configuration were only fixed at a late stage in the project - initially a T-tail and H-tail had been shortlisted for consideration by the Aerodynamics subteam; the H-tail won out to avoid the need for a very large vertical stabiliser. Throughout the design process, there were clearance issues with the rear pair of vertical rotors, as well as with the pusher rotor, leading to a twin-boom tail being considered as well. Although this would have decoupled tail placement from fuselage length, there would have been strength issues with the pylons having to support very large moments. It was also advantageous, both aesthetically and aerodynamically, to lengthen the fuselage and taper it. Although this would come with a weight penalty, it was also necessary to place the tail further back to obtain a reasonable static margin, as calculated by Control. Accordingly, the fuselage was lengthened beyond the necessary length found for packing concerns to allow for the original H-tail design to be attached to the end. The pusher rotor was placed behind the H-tail with minimal clearance, allowing the fuselage to be as long and streamlined as possible within the 1.5 m limit. This necessitated a partially-outboard driveshaft, however, the structural and weight complications of integrating the pusher between the tail and fuselage outweighed the advantages of a fully-enclosed system. Moreover, the blockage effects of the fuselage on the rotor would have been undesirable. The twin rotors were moved far enough outboard on the wing by Structures to clear the tail (aided by the resized propellers).

The mission profile also varied significantly throughout the design of the UAV. Depending on the

payload drop method and the respective drop height, different optimisations for the profile were possible. Control wanted a loiter and drop height as high as possible to maximise search area and minimise time to drop, but the drop accuracy for the unguided payloads was a constraining factor. This led to a maximum drop height being found as discussed in Section 5. However, as this was insufficiently high for search purposes, a different loiter height was established by Performance and the two phases were decoupled from each other. The minimum safe altitude (MSA) for takeoff was calculated to be 6 m, based on a 4 m wave height corresponding to the highest mean wind speed in the Mediterranean of 12 m/s (Force 6 winds [3]) and multiplied by a safety factor of 1.5. To avoid hitting any man-made obstacles, it was decided that the MSA would be measured off the launch ship’s deck to allow for clearances on large ships.

There were concerns raised by Performance towards the end that the aircraft’s phugoid mode damping ratio was insufficiently large in open-loop. These concerns were disregarded, as the drone is expected to be fully-autonomous, and hence in a closed-loop system, at all times. As the aircraft was still technically stable in open-loop, this was deemed acceptable and no physical design changes were made.

Finally, the placement of the quad-rotors was contested: placing them on top of the motors (and above the wing) would cause unwanted interactions between them and the wing during the transition phase, reducing the lifting efficiency of the wing, but give better clearances to obstacles and slightly reduce the effect of blockage from the pylons; placing them below would have prevented the rotor wake from affecting the wing, but would have made the rotors more susceptible to damage and blockage. Aerodynamics gave a figure of 5% blockage for each rotor from the pylon, but as the battery was sized to the limit of being able to vertically take-off under maximum load, it was decided that preventing loss of VTOL thrust was more important than reducing rotor wake effects. Thus, the rotors were ultimately placed above the electric motors.

3 Regulations

An important consideration was the extent of regulations for the aircraft in the target market(s), for the purposes of facilitating sales and its usage in those regions. The Mediterranean was the main area of concern, with an especially high number of migrant deaths on routes from north Africa to Italy.

In general, there are few regulations for unmanned aerial vehicles as they are relatively new to commercial aviation. This was reflected in the lack of regulations for UAVs from relevant north African countries, while even EASA only had skeleton regulations for UAVs with MTOMs of over 25 kg. The vast majority of larger UAVs are individually certified, with little regularity.

If the aircraft were human-operated, it would have fallen into the “Certified”, C6 category under Chapter III, Article 40 of EASA UAS regulations [4]. Despite this UAV’s design to have been fully autonomous, the spirit of the regulations were followed as much as possible. However, most of the stipulations related to the operations and registration of the UAV, rather than actual performance requirements. It was obligatory to include a green flashing light on the drone for visual identification, and, as the drone would fly higher than 400 ft (the usual drone flight ceiling [4]), an ADS-B transponder to assist air traffic control. A constant telemetry link with the option for manual command-and-control takeover was also mandated, but already included in the design owing to the mission parameters. Finally, the UAV was also required to possess geo-awareness for areas such as no-fly zones, however, as an emergency services drone, these were deemed largely irrelevant and it would be possible to easily incorporate this capability anyway if needed for certification.

The regulations that did pertain to airworthiness were EU Regulations 2015/640, 1321/2014 and 748/2012 [5, 6, 7]. These stated that such a UAV should be certified to a standard similar to that of general aviation aircraft under CS-23 [8], but did not provide exact quantitative requirements either. Hence, certain limits and constraints (such as the maximum gust speed as discussed later) were taken from CS-23 to adhere as closely as possible to existing regulations.

4 Cost Estimation and Lifespan

To supplement the business cases analysed by the other technical director, a detailed cost breakdown was produced. The three major pillars of cost for the program were the parts and raw materials cost, manufacturing costs and mission-specific costs. A big factor for the overall program cost was the estimated production quantity - this impacted the bulk purchase cost of raw materials and had an especially drastic impact on manufacturing costs, as mass-manufacturing allows tooling to be reused and a learning curve effect to be applied.

An figure of 100 aircraft was chosen for the initial production quantity after analysis of the following factors: Firstly, to obtain a reasonable deployment frequency, data on case frequencies and concentrations was obtained by the other technical director - across the Mediterranean from 2014, Coast Guards averaged 0.56 missions per day. The target region for this UAV, with a coastline within the jurisdiction of the Algerian, Tunisian and Libyan Coast Guards, contained about 75% of those cases (Figure 1) - assuming that a singular Coast Guard force in that area took on about 25% of the total, this gave them about 50 sorties per year.

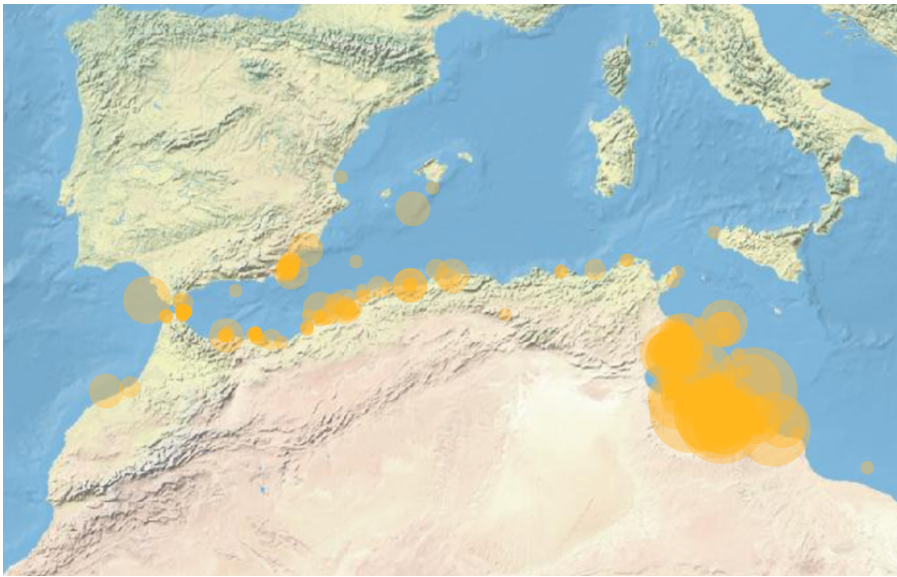


Figure 1: Migrant Casemap in the Mediterranean [1]

From the customers' side, typical Coast Guard budgets are 100-200 million GBP annually [9, 10, 11], which they then allocate 1-2% of to new projects. This translated to approximately 1-2 million GBP per customer on average. Figure 2 shows that at 100 aircraft, the price per UAV sits at around 150,000 GBP. Coupled with this cost vs. quantity analysis, we expect to have our top three target customers, the three Coast Guards previously mentioned, purchase approximately 10 drones each - a target mission expectancy of 80 per drone then gives them a total program lifespan of almost 16 years, which is within the typical military drone service life of 15-20 years [12].

Coast Guards typically spend about 10,000 GBP per sortie for surface vessels, and 30,000 GBP per helicopter mission [13, 14]. Use of this UAV also requires the deployment of a surface vessel to complete the rescue - the aim of the aircraft was to provide a cost-effective alternative to helicopter missions while providing nearly the same life-saving capabilities. In order to break even against using helicopter missions, each drone would have to complete a minimum of 8 sorties of approximately 1 flight hour each; this was obtained by dividing the total per-aircraft lifetime cost (mission-specific costs + initial procurement cost) by the number of missions and adding on the cost of a Coast Guard boat mission. Thus, an 80-mission service expectancy amounts to 10 times the minimum mission requirement; if the drone meets this expectancy, the overall per-mission cost was projected to be approximately 2250 GBP, yielding a total rescue cost of 12250 GBP - less than half the cost of a helicopter sortie.

In general, fatigue was not expected to be an issue owing to the low number of expected flight hours of the drone, and fatigue analyses performed by the Structures subteam confirmed this. Rather, the biggest threat to the operation of the drone was catastrophic failure caused by extreme weather conditions. The global failure rate per flight hour from systems failures specified in CS 23 was 1 in 10,000 [8] - however, the failure rate for this UAV is likely to be far higher than that owing to its smaller size and expected deployment conditions. Data was extremely scarce for failure rates related to large UAVs in rough weather, but a study suggested that wind speeds in the Mediterranean only exceed 25 m/s (48.5 kts or Force 10 [3]) once every 30 years on average [15], implying that the drone will be able to survive most conditions in the Mediterranean. The drone was designed to operate optimally in Force 6 winds or below - although winds of between Force 6 and 10 are likely to be more common, the same study also suggested that these high wind speeds are usually caused by the passage of specific storms which are rare, rather than a part of the regular climate.

Overall, producing 100 aircraft would cause our target market to represent about 30-35% of sales. 10-15% would be allocated to spares and/or additional purchases from the top 3 customers. The remaining 50-55% allows for expansion to other markets, such as Coast Guards in the Caribbean, as well as other emergency services or private buyers (discussed further by the other technical director). This number constitutes a good compromise between expected sales and production cost, allowing the use of batch production techniques to cut costs. Thus, the entire program would cost approximately 15 million GBP.

Compared to initial procurement costs for surface vessels, this program requires far less initial investment: coastal patrol vessels cost in the single-digit millions of pounds per boat [16, 17, 18], while larger offshore patrol vessels can cost over 10 times that in the hundred-million pound range [19, 20]. A fully-operational fleet of 10 UAVs could be purchased for less than 2 million GBP.

Additionally, there is potential for relatively easy expansion of the program if more sales interest is garnered as tooling is likely able to be reused for further batch production, with the procurement cost per aircraft dipping below 100,000 GBP each at around the 300 production quantity mark.

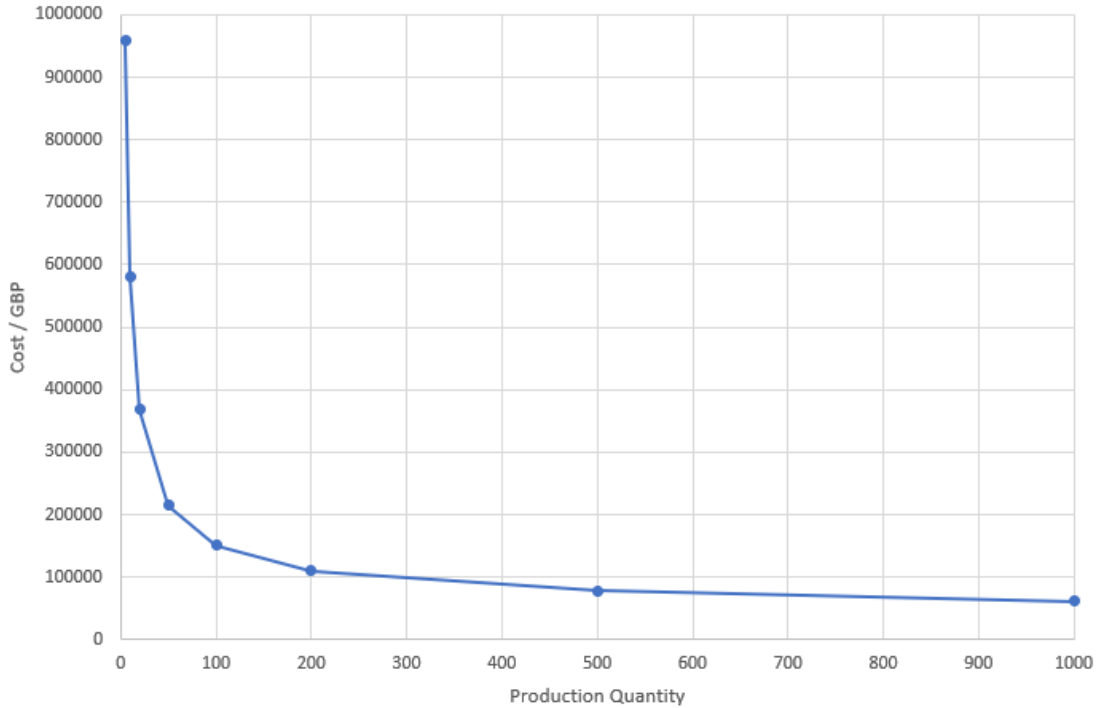


Figure 2: Cost vs. Quantity

4.1 Parts

An in-depth list of parts for the avionics and propulsion systems was provided by the Propulsion and Systems team, whom in most cases were also able to provide the cost of the item. A flight control computer, a battery management system, telemetry, various necessary sensors, actuators and wiring were included in the avionics, while a fuel tank, a battery pack, electronic speed controllers, an internal combustion engine, electric motors, a clutch, an APU and the propellers made up the cost of the propulsion systems. Some parts, being relatively new, such as Honeywell’s satellite communications unit [21], did not have a price quote readily available - hence the pricing was estimated from other comparable components found on the market.

Likewise, Structures provided the total masses and types of raw material needed to construct the fuselage, wings and empennage for each aircraft; this was then used in conjunction with the expected production quantity to estimate the bulk purchase cost of the material. The manufacturing methods chosen were mainly press-moulding for most structural components, which has a 100% material efficiency, requiring no corrective factor for the cost. Similarly, composite layups also have a very high material efficiency. A small amount of subtractive manufacturing is required on the aircraft, but this was taken into account with the cost rounding. A breakdown of the parts cost is provided in Table 1, with the costs rounded up to the nearest 100 GBP:

Table 1: Part costs

Part	Price / GBP
Avionics	7000
Thermal Camera	5500
Propulsion	3700
Structural material	200
Batteries	2000

This gave a total parts cost of 18,500 GBP per UAV.

4.2 Manufacturing

The manufacturing costs were by far the largest portion of the total program cost, comprising over 80% of it. The costs were broken down into different phases: tooling, manufacturing, quality control, development support and flight testing. Empirical relations based on the MTOM and maximum speed of the aircraft were used to estimate the number of man-hours for each stage, which was then multiplied by current employee “wrap rates” (salaries + benefits + insurance etc.) to find the total cost [22]. As the aircraft would most likely be batch produced rather than mass-manufactured, process automation was assumed not to be cost-effective. Two flight-testing aircraft were costed for, reflecting the lower certification requirements for UAVs. Table 2 shows the breakdown:

Table 2: Manufacturing costs

Phase	Price / GBP
Tooling	2,632,544
Manufacturing	6,913,039
Quality Control	1,013,254
Development Support	843,303
Flight Testing	1,314,661

This yields a total manufacturing cost of about 13.25 million GBP, or 132,500 GBP per UAV.

4.3 Mission-specific

The main mission-specific costs ensued from fuel, maintenance (parts and labour) and the payload itself. The UAV carries 4.5 kg of 87-93 Octane gasoline per mission (for which reference prices were taken from Italy [23]), essentially a negligible cost.

Rather, the price of each life-saving payload can be significant, especially if they are not reusable or re-packable. Although the aircraft does not include a payload on delivery, the reference liferaft that was designed for costs approximately 2000 GBP [24].

Furthermore, some maintenance procedures for the drone will be required after every mission. Although the drone was designed to be low-maintenance, it will be necessary to re-pack or replace the payload manually, as well as refuel and recharge the drone and repack the airbags after landing. This sums up to approximately 1.2 maintenance man-hours per flight hour on average, split between scheduled maintenance (once every 10 sorties [25]) and mission-critical maintenance (replacing payloads). The maintenance parts cost was calculated as 1/80th (from mission expectancy) of the total initial parts cost. Table 3 shows the cost of maintenance per mission:

Table 3: Mission-specific costs

Item	Price / GBP
Maintenance (labour)	150
Parts	230
Fuel	10

Hence, the overall per-aircraft lifetime cost (assuming that they meet the mission expectancy) would be 181,000 GBP, excluding payload costs.

5 Payload Drop Method Trade-off Analysis

The initial mission profile for the UAV called for a stationary payload drop in hover mode. However, upon further analysis, it was discovered that there were certain efficiency and accuracy gains to be had by utilising different drop methods. There were four drop methods initially considered:

1. Stationary hover drop
2. Low-speed forward-moving drop using the electric motors
3. Drop in steady, level flight at stall speed
4. Drop in a steep dive

The aim was to determine, firstly, whether methods 3 and 4 were feasible and provided better accuracy than methods 1 and 2, or, failing that, the ideal height and airspeed to drop the payload from.

In order to accurately model the different payload trajectories, a first-order iterative method that accounted for aerodynamic drag was used [26]. A timestep of 0.01 s was taken, with a tolerance of 0.01 m on the height variable to ascertain when the payload had reached the surface. Six main factors affecting the payload's trajectory were considered: the ground speed of the aircraft, the wind speed and direction, the gust speed and direction and the release delay from the payload drop mechanism. A multi-directional pitot tube system was incorporated so that the aircraft would be able to measure the wind speed in real-time, with variations taken into account by the gust factor.

In keeping with the goal of optimising the payload drop for accuracy, the maximum possible deviation from the target for different ground speeds were calculated. The target was the projected point of landing where initial ground speed and windspeed were taken into account, while the "actual" landing position of the payload was calculated by modelling an additional gust factor and the release

mechanism delay. A 5 m radius for maximum deviation was imposed, ensuring that the average human was within approximately three armspans' length [27] of reaching the life-saving device. The highest mean windspeed occurring in our main target region between southern Sardinia and north Africa of 12 m/s was taken as the maximum [28], equivalent to Force 6 on the Beaufort Wind Scale [3]. To account for the biggest possible deviation of the payload, a maximum horizontal gust velocity of 7.62 m/s was used throughout. In the spirit of the following existing regulations where current ones do not exist for UAVs, this corresponds to the requirement of 25 ft/s stipulated by CS-23 [8]. Thus, a maximum drop height to hit the 5 m criterion was obtained for each different ground speed.

The aerodynamic drag on the payload was calculated using the following equation:

$$D = \frac{1}{2} C_D \rho U^2 A \quad (2)$$

Using equation 2, the acceleration of the payload in each direction was calculated and its ballistic trajectory estimated by iterating over the position, velocity and acceleration of the payload until its height reached zero. Wind directions were discretised into three directions, namely forwards, backwards and sideways, with the sideways deviation being symmetrical. Although this was not representative of all possible target deviations by the payload, maximum deviation would only occur if the aircraft's ground speed and the wind and gust directions were exactly aligned or opposed, so it was an appropriate assumption to make. Sideways deviations were checked but always remained well within the 5 m limit, proving irrelevant to the analysis.

The drag coefficients were taken from empirical data [29, 30] and based on the different cross-sectional shapes seen by each particular wind direction. The payload was assumed to drop perfectly in the orientation that it would be stored in the aircraft - with freefall times of approximately two seconds, it was unlikely that the payload would rotate significantly. The drop time of the payload is entirely dependent on the drop height, with higher values allowing more time for the payload to be affected by the wind and errant gusts. Both the liferaft and lifebuoy payloads were modelled. Table 4 shows the drag coefficients for each case:

Table 4: Payload Drag Coefficients

Payload	Cross-section	C_D
Liferaft	Bottom	1.28
	Front/Rear	0.82
	Side	1.15
Lifebuoys	Bottom	0.82
	Front/Rear	0.74
	Side	0.74

The bottom of the liferaft payload was modelled as a flat plate, the front/rear as a long cylinder with axial airflow and the side as a short cylinder. The bottom of the lifebuoy canister (dropped axially vertical) was modelled as a long cylinder, and the front/rear/sides as a cylinder encountering radial airflow. These C_D s were coupled with the respective cross-sectional areas to provide an accurate drag estimation. As the drop height was very low, variations in air density were considered negligible and taken at the sea level value of 1.225 kg/m³.

It was discovered from the beginning that dropping the payload into a headwind would drastically increase accuracy, as the speed of the drone would somewhat counteract the wind's effects. Thus, with this being stipulated as a path-planning constraint, only the maximum deviation from this condition was considered. Although the MSA was 6 m, the initial drop height was taken to be 15 + 6 m - this was to account for vertical gusts (maximum of 50 ft/s from CS-23 [8]), while still providing minimal payload freefall time. The release delay, although not easily quantifiable as it would be system-specific depending on the particular electronic components used, was taken to be 0.2 s. This time was found to be the typical 60° rotation time of a high-torque servo [31], representing the operation time of the release mechanism.

Figure 3 shows the maximum deviation of Method 3, varying with ground speed with the parameters specified above. The drop ground speed for this case was varied between 18 and 26 m/s, with 18 m/s being the minimum ground speed required not to stall with a maximum wind speed of 12 m/s, and 30 m/s the groundspeed required if there was no wind.

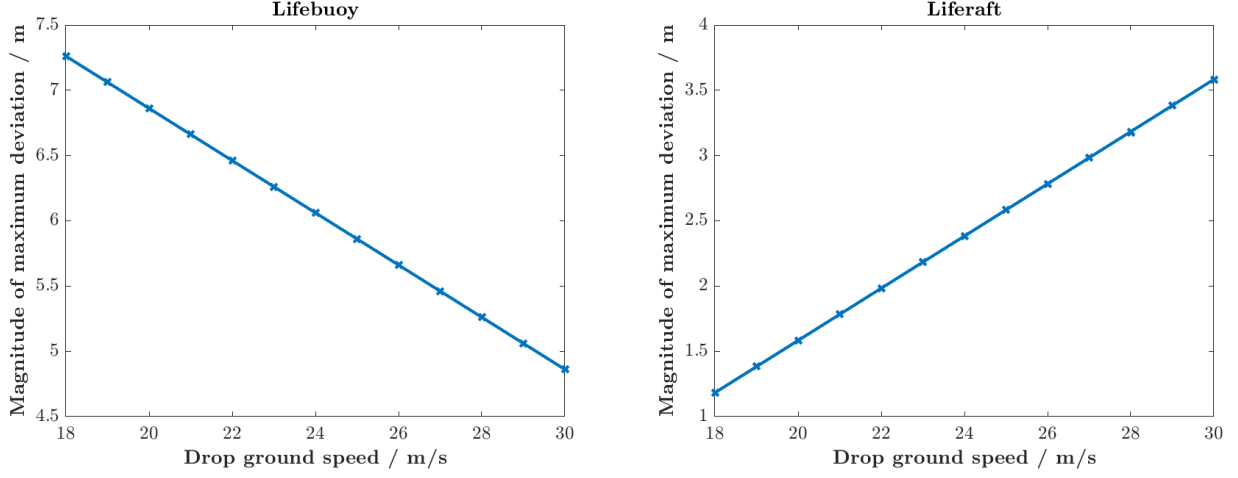


Figure 3: Method 3

Although the liferaft has an acceptable deviation for all cases, it can be seen that the lifebuoy canister did not satisfy the accuracy requirement for all but the “no wind” case. This meant that Method 3, in spite of being the most efficient energy-wise, had to be discarded, and Method 4 was then analysed in search of potential accuracy gains.

The limiting factor for the dive-bomb method was the minimum pull-up radius of the aircraft, constrained by the maximum airframe load factor of 3. The aircraft was assumed to be in a near-vertical dive when dropping the payload, so the drag coefficients and reference areas were changed according to the 90° rotation in payload orientation. Starting from the loiter velocity of 41.3 m/s and factoring in freefall time, Performance provided the minimum pull-up height of 248 m for a load factor of 2.29, which was added to the MSA of 6 m to get the payload drop height of 254 m. The increased minimum stall speed at this load factor was accounted for as the starting drop speed of the payload. Figure 4 shows the respective deviations, as before:

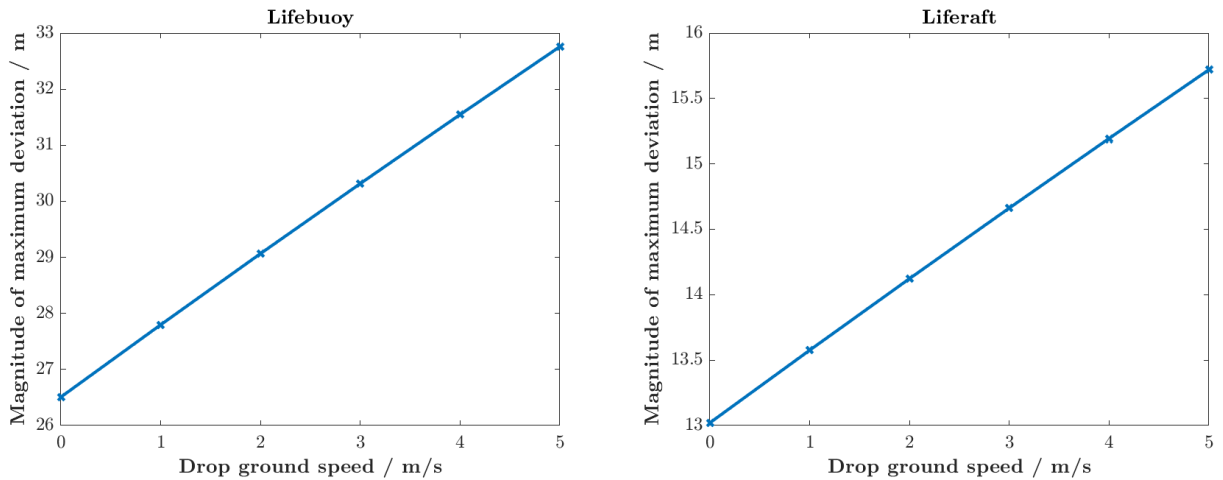


Figure 4: Method 4

In this scenario, both the deviations of the liferaft and lifebuoy canister payloads far exceeded the 5 m requirement - the airframe was not designed to resist high g-forces, leading to poor performances in this case. Additionally, there were safety concerns with extremely high payload impact speeds,

as well as complications with the payload release mechanism in a vertical dive.

Consequently, Methods 1 and 2 were analysed for accuracy. A range of different drop heights were tried and the maximum deviation at each noted. Method 1, being the hover drop mode, was analysed with a windspeed range from 0 - 12 m/s. Method 2 was analysed with a constant aircraft airspeed of 12 m/s; the wind and ground speeds were varied complementarily to match, i.e. at 10 m/s ground speed the wind speed would have been 2 m/s. Figures 5 and 6 show the deviations for these methods at a drop height of 18 m - this was found to be the maximum height where the largest deviation did not exceed 5 m. Even though Method 1 was more accurate for both payloads at 18 m, Method 2 still satisfied the 5 m constraint at the same height, and was more energy efficient through the use of Effective Translational Lift (ETL). Crucially, it was also impossible for Method 1 to be employed at a higher drop height, as the hover mode's inability to counteract strong winds meant that its accuracy was already on the limit at the top end of the wind speed range.

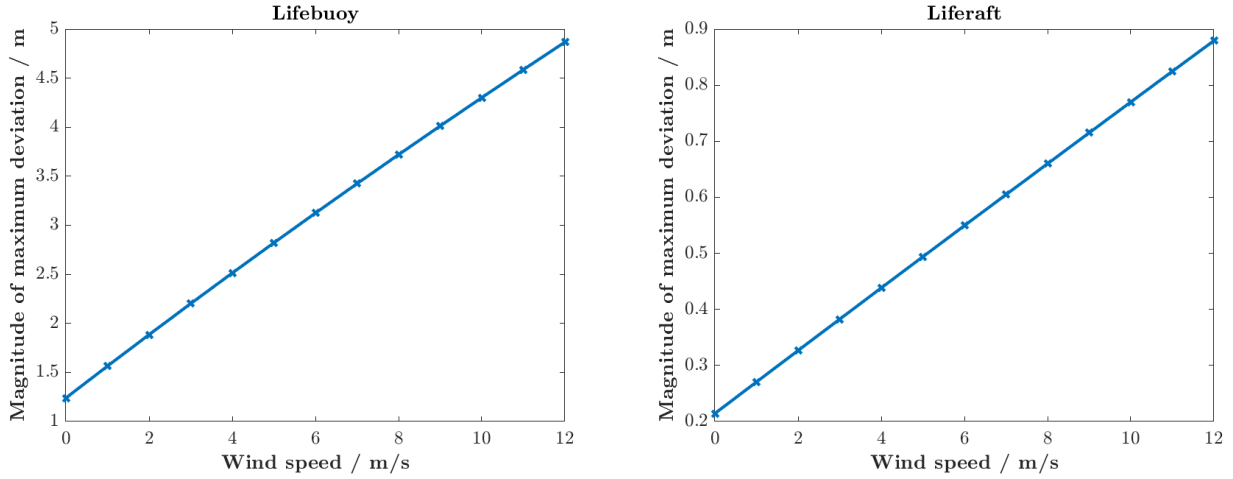


Figure 5: Method 1

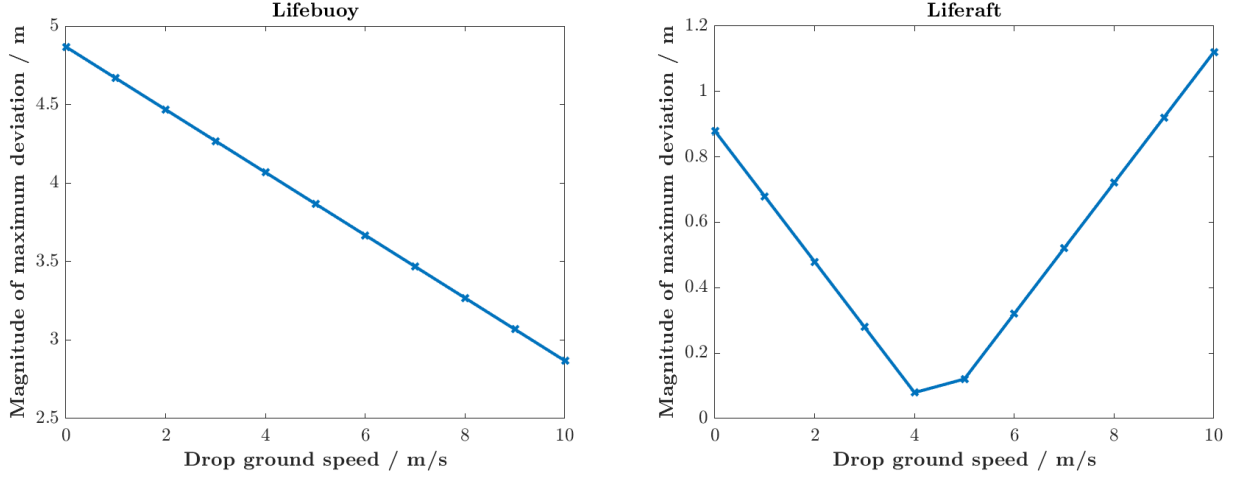


Figure 6: Method 2

As such, with no obvious disadvantages to using Method 2 over Method 1, the former was selected as the payload drop method. It was noted that, depending on the accuracy requirements, the drop method could be modified for other use cases if either hover drops or level-flight drops were specifically required.

5.1 Evaluation

There were two major sources of error in the analysis: 1. The assumption that the payload orientation would be perfectly horizontal (vertical for the dive-bomb case) and 2. The drag coefficients taken from literature.

It is unlikely that the aircraft will be completely horizontal when the payload is dropped, as it would need to tilt forwards to move forwards in VTOL mode. This would affect the relevant drag coefficients of the payload, as well as changing the cross-sectional area presented to each wind direction. Moreover, the drag coefficients found were estimates from empirical relations that did not necessarily match the conditions the payload would experience. More accurate models could be produced by obtaining the exact drop angle from Propulsion and calculating the requisite cross-sectional areas. Wind-tunnel testing could be employed with representative cylinders or boxes to find accurate drag coefficients for the payloads from each wind direction.

6 Conclusion

At the conclusion of the project, a fully-autonomous, VTOL-capable fixed-wing UAV, able to carry up to 9 kg of life-saving payload was successfully designed by the FLR22 team. Fregata adhered to existing EASA CS-23 and UAV regulations as much as possible. It was successfully able to augment Coast Guard operations by providing a rapid and cost-effective alternative to helicopter sorties in gale-force weather, with a projected payload delivery accuracy of 5 m and a maximum response time of 30 minutes to a target anywhere within the Mediterranean when used in conjunction with relevant Coast Guard boats.

References

- [1] MIGRATION WITHIN THE MEDITERRANEAN, 2022.
- [2] Michael Volker. Cold Water Survival, 7 2021.
- [3] Francis Beaufort. Beaufort Wind Scale, 1805.
- [4] Easy Access Rules for Unmanned Aircraft Systems (Regulation (EU) 2019/947 and Regulation (EU) 2019/945), 9 2021.
- [5] EASA. COMMISSION REGULATION (EU) No 2015/640. Technical report, EASA, 4 2015.
- [6] EASA. COMMISSION REGULATION (EU) No 1321/2014. Technical report, EASA, 11 2014.
- [7] EASA. COMMISSION REGULATION (EU) No 748/2012. Technical report, EASA, 8 2012.
- [8] EASA. Easy Access Rules for Normal-Category Aeroplanes (CS-23) (CS Amendment 5, AMC/GM Issue 2). Technical report, EASA, 2019.
- [9] Charity Commission for England and Wales. THE ROYAL NATIONAL LIFEBOAT INSTITUTION, 12 2020.
- [10] Department of Transport. Budget 2022 to help transform how we travel, 10 2021.
- [11] Running costs, 2020.
- [12] Zachary Morris. Smaller, Less Capable Drones for the Near Future, 6 2018.
- [13] Dan Parsons. Irish Coast Guard shopping for new search-and-rescue service, 1 2022.
- [14] Bristow Search and Rescue: UK Operations, 2022.
- [15] Stephen Outten and Stefan Sobolowski. Extreme wind projections over Europe from the Euro-CORDEX regional climate models. *Weather and Climate Extremes*, 33, 9 2021.
- [16] Algeria getting more FPB 98 patrol boats, 2 2019.
- [17] Yusuf ÇETINER. OCEA Launches First FPB 98 Patrol Boat Built For Ukraine, 12 2021.
- [18] George Tsiboukis. New coastal patrol boats for the Hellenic Coast Guard, 3 2019.
- [19] Offshore patrol boat arrives in home port, 7 2010.
- [20] India approves plan for \$700 million acquisition of 6 next-generation offshore patrol vessels, 8 2018.
- [21] THE MOST RELIABLE SATCOM SOLUTION FOR UAV, 2022.
- [22] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 6 edition, 2018.
- [23] Italy Gasoline prices, 13-Jun-2022, 6 2022.
- [24] Aero Compact 4 Man Liferaft, 2022.
- [25] Routine Drone Maintenance Checklist, 2022.
- [26] Rohan Pratap Singh and Akash Garg. Autonomous payload drop system using mini Unmanned Aerial Vehicles. Technical report, 2016.
- [27] Amitav Sarma, Bhupen Barman, GautamC Das, Hiranya Saikia, and AmbathD Momin. Correlation between the arm-span and the standing height among males and females of the Khasi tribal population of Meghalaya state of North-Eastern India. *Journal of Family Medicine and Primary Care*, 9(12):6125, 2020.
- [28] I. Koletsis, V. Kotroni, K. Lagouvardos, and T. Soukissian. Assessment of offshore wind speed and power potential over the Mediterranean and the Black Seas under future climate changes, 7 2016.
- [29] Nancy Hall. Shape Effects on Drag, 5 2021.
- [30] C F Heddleson and D L Brown. B*4 Summary of (Drag Coefficients of (Various Shaped Cylinders. Technical report.
- [31] HS-645MG High Torque, Metal Gear Premium Sport Servo, 2022.

Appendix

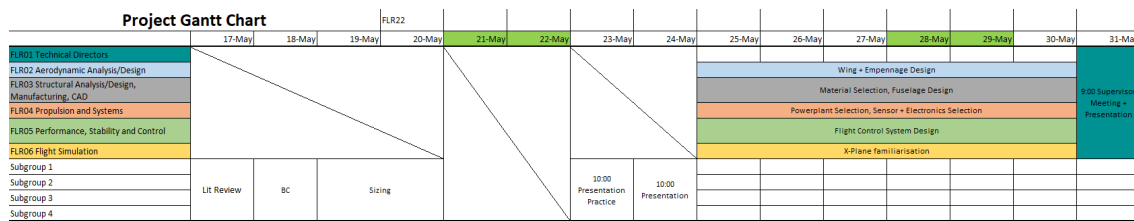


Figure 7: Gantt Chart excerpt

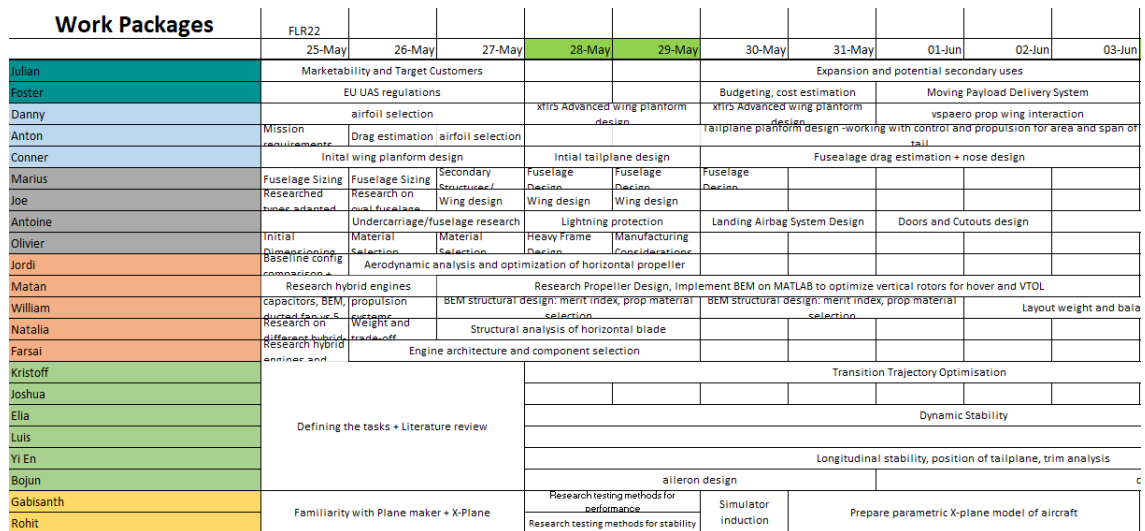


Figure 8: Work Breakdown excerpt